Fangshuo Xu

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Education

Hunan University Sept. 2023 – June 2027 (Expected)
School of Artificial Intelligence and Robotics
B.Eng. in Robotics Engineering
GPA: 3.8/4.0 Rank: 5/71 (Top 7%)
Compulsory courses: Principle and System of Signal Processing (95), Computer Composition Principle and Embedded System (94), Circuit Electronics (94), Linear Algebra A (93), Basic Physics of Robot (91), Intermediate Practice of Robotics Engineering (96), Control Principle and Control System (88)

Publication

Wang, N.[1], **Xu, F.[1]**, Dian, R.[*], Li, S. (2026). *Imaging-Informed Progressive Fusion for Mosaic-based Hyperspectral Image Reconstruction*. Submitted to *Information Fusion*.

[1]Equal contribution.

Projects

Imaging-Informed Progressive Fusion for Mosaic-based Hyperspectral Image Reconstruction Mar. 2026 – May 2026

- Achieved **41.84 dB PSNR** and **0.9880 SSIM** on the CAVE dataset, outperforming EBVIF by **+0.92 dB** in PSNR through an unsupervised framework for mosaiced and PAN image fusion.
- Designed a physics-guided progressive architecture that aligns hierarchical sampling and average-pooling degradations with corresponding inverse reconstruction stages, bridging physical imaging mechanisms with deep reconstruction.
- Achieved state-of-the-art real-world performance with **0.9186 QNR** by introducing cross-scale parameter sharing and a three-stage composite loss to improve interpretability, structural consistency, and reconstruction fidelity.

Stereo Visual Odometry for UAV Navigation Sept. 2025 – Dec. 2025

- Implemented a stereo VO pipeline combining LK optical-flow tracking, stereo triangulation, and PnP/RANSAC-based relative pose estimation for GPS-denied localization.
- Improved VO robustness by adding motion prediction, low-pass velocity filtering, outlier rejection, reprojection-error filtering, and an inertial coasting mechanism during temporary feature loss.
- Integrated VO outputs into a UAV navigation workflow for downstream planning, control, and RViz visualization, supporting OptiTrack-assisted real-world UAV flight testing in GPS-denied environments.

Auto-Aiming Platform (2025 National College Students' Electronic Design Contest) July 2025 – Aug. 2025

- Achieved stable real-time target localization at **>25 FPS** by deploying YOLOv5-based black-border detection on MaixCAM and restricting subsequent processing to ROI crops.
- Developed a geometric correction pipeline using adaptive thresholding, flood filling, contour approximation, and perspective transformation to recover the paper target center under angular distortion.
- Generated third-circle contour points and transmitted rectified center coordinates through serial communication, enabling downstream closed-loop aiming control based on image-center error.

Honors

Provincial Third Prize, National Electronic Design Contest (Hunan Province) (Among 874 teams) 2025
First-Class Academic Scholarship, Hunan University (Major Top 6%) 2024

Skills

Programming: C/C++, Python, MATLAB
Robotics: ROS, PX4, MAVROS, RViz, Linux, Eigen, PCL
Computer Vision & AI: OpenCV, PyTorch, YOLOv5, Visual Odometry
Tools & Hardware: Git, LaTeX, Intel RealSense, OptiTrack